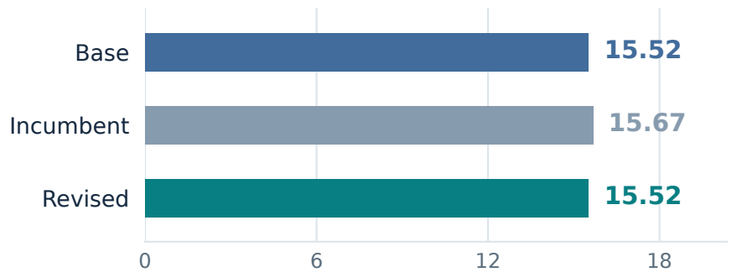
# Signed arithmetic coverage repairs the remaining panel regressions

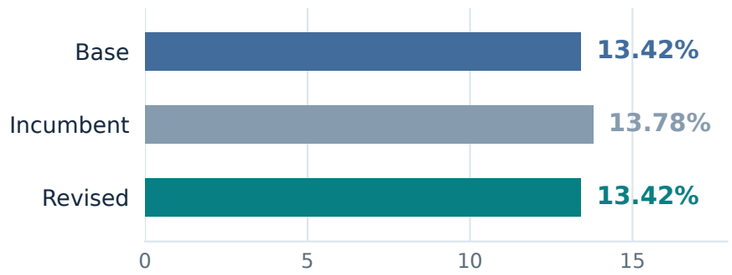
32 new training circuits · one fixed 300-tree fit · input-only signed repair route · base cap decisions preserved

## A  Consumed arithmetic stress · 32

MAE · LUT4 · lower is better



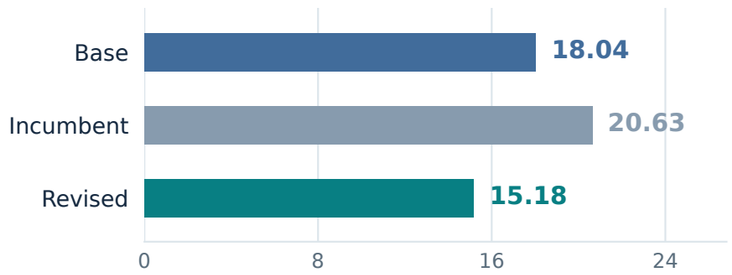

Positive-target MAPE · lower is better



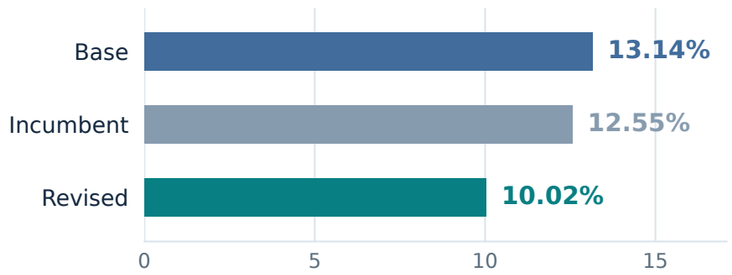

**Both named regressions fixed; all six retained panels have MAE and MAPE no worse than base.**

## B  Consumed signed dual arithmetic · 4

MAE · LUT4 · lower is better

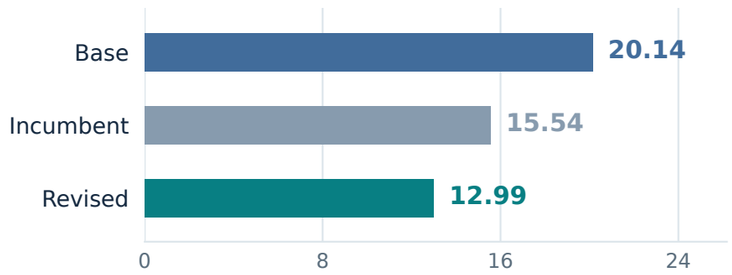



Positive-target MAPE · lower is better

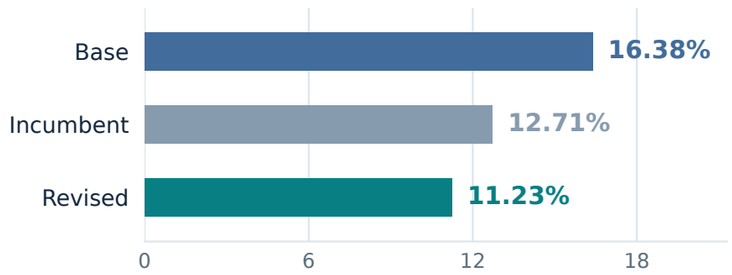



## C  Independent fresh check · 40

MAE · LUT4 · lower is better




Positive-target MAPE · lower is better




**Fresh and development gates passed.**

## Fresh budget mistakes are unchanged for all three predictors

| Cap · LUT4 | False accept | False reject | Predicted / actual feasible |
|---|---|---|---|
| **32** | **0** | **0** | **6 / 6** |
| **64** | **0** | **0** | **10 / 10** |
| **128** | **2** | **0** | **27 / 25** |

## Worst underprediction · LUT4

**83.09 → 80.02 → 80.02**

Base → incumbent → revised

Exact decisions at caps 32 / 64 / 128;

**the two false accepts at 128 remain.**

Visible tradeoffs: fresh signed-variable-add/boolean subgroup (4 cases) MAE 11.67 → 12.55 vs base; MAPE 11.42% → 10.01%.

Retained research MAE 31.60 → 31.63 vs incumbent (base 32.16). No new subgroup is an independent population guarantee.

The predeclared static-signed fallback was selected on consumed development data; the primary missed its retained research MAPE gate.

All earlier 192 / 327 / 32 / 40 / 40 / 32 panels are consumed. The fresh 40-case cohort was frozen separately; no learning curve is implied.

Experimental isolated iCE40 LUT4 point estimates only. No calibrated safety, full-CPU, timing, power or application-transfer guarantee. Base stays the default.